



# EDR

## Endpoint Detection & Response

Architecture, Workflow & SOC Management

# AGENDA

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# WHAT IS EDR?

**EDR** (Endpoint Detection & Response) is a cybersecurity solution that continuously monitors endpoint devices—laptops, servers, workstations, and mobile devices—to **detect, investigate, and respond to threats in real time.**



## Continuous Monitoring

Records all endpoint activity 24/7 — processes, network connections, file changes, registry modifications.



## Real-Time Detection

Uses behavioral analytics, ML, and threat intel to identify malicious activity as it happens.



## Automated Response

Isolates compromised hosts, kills processes, and blocks malicious files automatically.



## Forensic Investigation

Provides full attack timeline and root cause analysis for every incident.

# EDR ARCHITECTURE

## LAYER 1 – ENDPOINTS

Laptops / Desktops

Servers

Mobile Devices

Virtual Machines

Cloud Workloads

## LAYER 2 – EDR AGENT

Kernel-level Hooks

Process Telemetry

Network Monitoring

File I/O Capture

Registry Events

## LAYER 3 – EDR PLATFORM / CLOUD

Behavioral Engine

Threat Intelligence

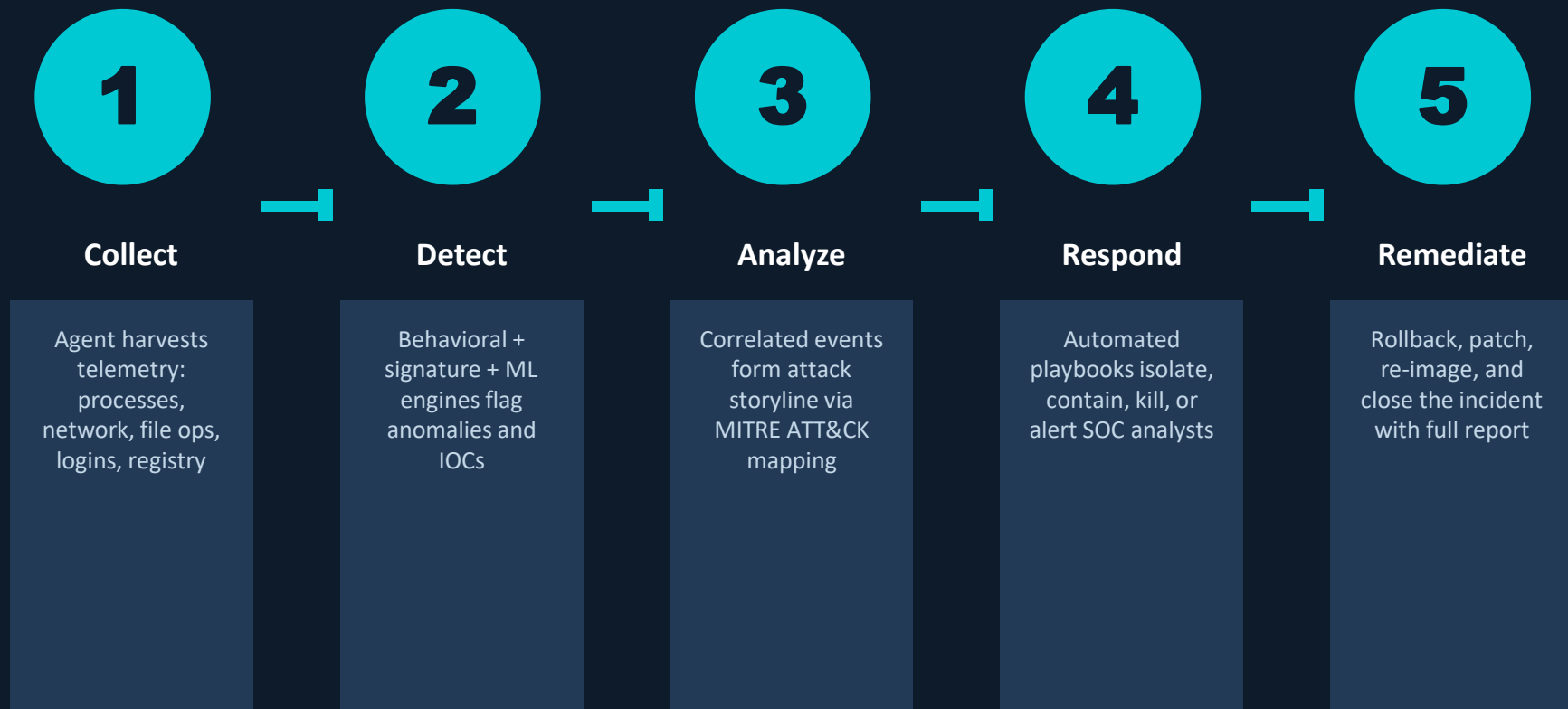
ML Analytics

Alert Management

SOAR Integration

*Data flows upward: Agent captures endpoint telemetry → forwards to EDR Platform for analysis → triggers automated or analyst-driven response*

# HOW EDR WORKS



# EDR IN SOC MANAGEMENT

## Tier 1

### Alert Triage Analyst

- ▶ Monitor EDR alert queue
- ▶ Classify: true positive vs. false positive
- ▶ Initial scope assessment
- ▶ Escalate high-severity alerts to Tier 2

## Tier 2

### Incident Responder

- ▶ Deep-dive investigation using EDR timeline
- ▶ MITRE ATT&CK technique mapping
- ▶ Containment & isolation via EDR console
- ▶ Evidence collection for forensics

## Tier 3

### Threat Hunter / IR Lead

- ▶ Proactive threat hunting across fleet
- ▶ Custom detection rule authoring
- ▶ Root cause analysis & attribution
- ▶ Post-incident playbook improvement

Escalation Flow →

# SOC EDR WORKFLOW & PLAYBOOK



# KEY EDR CAPABILITIES



## Threat Hunting

Proactively search for hidden threats using IOH queries across the entire endpoint fleet.



## MITRE ATT&CK Mapping

Every alert tagged with tactic and technique IDs for structured analysis.



## Automated Playbooks

SOAR-integrated rules auto-contain, notify, and remediate common threat scenarios.



## Behavioral Analytics

ML models baseline normal activity and flag deviations — catches zero-days.



## Forensic Timeline

Full attack reconstruction: who, what, when, where — process tree to network callout.



## Cloud & Hybrid Coverage

Single pane of glass for on-prem, cloud VMs, containers, and remote endpoints.



# EDR vs TRADITIONAL ANTIVIRUS

|                  | TRADITIONAL AV         | EDR                                     |
|------------------|------------------------|-----------------------------------------|
| Detection Method | ✗ Signature-based only | ✓ Behavioral + ML + Signatures          |
| Zero-day Threats | ✗ Limited / Missed     | ✓ Strong detection capability           |
| Visibility       | ✗ File-level only      | ✓ Full process, network, memory         |
| Response         | ✗ Quarantine file      | ✓ Isolate host, kill process, rollback  |
| Investigation    | ✗ Not available        | ✓ Full forensic timeline & process tree |
| Threat Hunting   | ✗ Not supported        | ✓ Active hunting with query language    |
| MITRE ATT&CK     | ✗ Not mapped           | ✓ Every alert mapped to ATT&CK          |

## KEY TAKEAWAYS

# EDR is the backbone of modern endpoint security

- EDR provides continuous, 24/7 visibility into every endpoint across your environment.
- SOC teams leverage EDR for structured alert triage, investigation, and incident response.
- Automated playbooks reduce MTTD and MTTR dramatically versus manual processes.
- MITRE ATT&CK integration enables precise threat classification and hunting.
- EDR is not a replacement for AV — it is a force multiplier for the entire SOC.

**Secure Every Endpoint. Respond with Confidence. Protect the Mission.**